United States Patent	12403619
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Holmes; Lonnie Andrew

Hair clippers with flexing electrically adjustable blades

Abstract

A flex clipper provides a feature to help the cutting blade set float more effortlessly by adjusting automatically to the contours of a client's head to prevent getting stuck and causing cuts and irritation to the scalp. The hair clipper preferably also uses a self-contained motor-driven pivotable adjustment mechanism to adjust the relative position of the stationary and reciprocating blades of a common type of blade set. One or more on/off switches operable by the thumb of the hand holding the clipper to afford a barber total automatic adjustment with the clipper itself in an on or off condition.

Inventors: **Holmes; Lonnie Andrew (Coram, NY)**

Applicant: **Holmes; Lonnie Andrew** (Coram, NY)

Family ID: **82323523**

Appl. No.: **17/546841**

Filed: **December 09, 2021**

Prior Publication Data

Document Identifier	Publication Date
US 20220219343 A1	Jul. 14, 2022

Related U.S. Application Data

continuation parent-doc US 15677018 20170815 US 10391646 20190827 child-doc US 16547535
continuation parent-doc US 14622554 20150213 US 9731424 20170815 child-doc US 15677018
continuation-in-part parent-doc US 16547535 20190821 US 11198229 child-doc US 17546841
continuation-in-part parent-doc US 13727274 20121226 US 9352476 20160531 child-doc US 14622554
division parent-doc US 12592537 20091124 US 8341846 20130101 child-doc US 13727274
us-provisional-application US 61117434 20081124

Publication Classification

Int. Cl.: **B26B19/04** (20060101); **B26B19/02** (20060101); **B26B19/20** (20060101)

U.S. Cl.:

CPC **B26B19/048** (20130101); **B26B19/02** (20130101); **B26B19/046** (20130101); **B26B19/20** (20130101); **B26B19/205** (20130101);

Field of Classification Search

CPC: B26B (19/02); B26B (19/048); B26B (19/063); B26B (19/205); B26B (19/046); B26B (19/20); B26B (19/04)

USPC: D28/44-54; 30/43.7- 4; 30/194-204; 30/208-220

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1516635	12/1923	Friedman	30/220	B26B 19/205
2858607	12/1957	Kane	30/43.6	B26B 19/16
2903789	12/1958	Schleifer	30/43.9	B26B 19/28
3287805	12/1965	Du Charme	30/202	B26B 19/20
6243955	12/2000	Forbers	30/195	B26B 19/20
7581319	12/2008	Little	30/210	B26B 19/06
8127453	12/2011	Haczek	30/43.92	B26B 19/048
8341846	12/2012	Holmes	30/208	B26B 19/20
9352476	12/2015	Holmes	N/A	B26B 19/20
9731424	12/2016	Holmes	N/A	B26B 19/02
10391646	12/2018	Holmes	N/A	B26B 19/063
11198229	12/2020	Holmes	N/A	B26B 19/20
11534932	12/2021	Johnson	N/A	B26B 19/20
11577414	12/2022	Buck, Jr.	N/A	B26B 19/063
2013/0326881	12/2012	Blatter	30/41	B26B 21/446
2022/0410420	12/2021	Reinprecht	N/A	B26B 19/20
2023/0077934	12/2022	Lin	30/539	B26B 19/046
2025/0135668	12/2024	Snow	N/A	B26B 19/205

Primary Examiner: Prone; Jason Daniel

Background/Summary

RELATED APPLICATIONS (1) This application is a continuation-in-part of application Ser. No. 16/547,535, filed on Aug. 21, 2019, now U.S. Pat. No. 11,198,229 B2, issued Dec. 14, 2021, which '535 application is a continuation of application Ser. No. 15/677,018, filed Aug. 15, 2017, now U.S. Pat. No. 10,391,646 B2 issued Aug. 27, 2019, which '018 application is a continuation of application Ser. No. 14/622,554 filed Feb. 13, 2015, now U.S. Pat. No. 9,731,424 issued Aug. 15, 2017, which '554 application is a continuation-in-part of application Ser. No. 13/727,274, filed on Dec. 26, 2012, now U.S. Pat. No. 9,352,476 issued May 31, 2016, which '274 application is a divisional of application Ser. No. 12/592,537, filed on Nov. 24, 2009, now U.S. Pat. No. 8,341,846 issued Jan. 1, 2013, which '018, '554, '274 and '537 applications are incorporated by reference herein. Applicant claims priority under 35 U.S.C. § 120 therefrom. application Ser. No. 12/592,537 is based upon provisional application Ser. No. 61/117,434 filed Nov. 24, 2008, which application is also incorporated by reference herein. Applicant claims priority under 35 USC § 119(e) therefrom.

FIELD OF THE INVENTION

(1) The present invention relates to hair cutting.

BACKGROUND OF THE INVENTION

(2) Electrically operated hair clippers have been used for many years. Some of the commonly available models have a manual lever on the side to incrementally adjust the relative position between the stationary and the reciprocating blades in a blade set to adjust the minimum length of hair that is being clipped. Other prior art patents show infinite adjustability over a range. The prior art does not reveal motor-powered continuous adjustability of the blade set which affords the barber the ability to perform the adjustment even during the clipping activity by simply activating a switch and/or having a flexing compliance blade set that adjusts around the contours of the scalp of a flex clipper is described which, in addition to the aforementioned powered hair cutting length adjustment feature, provides an additional feature to help the cutting blade set float more effortlessly by adjusting automatically to the contours of a client's head, to prevent the blade set getting stuck and causing cuts and irritation to the scalp of the customer.

OBJECTS OF THE INVENTION

(3) It is therefore an object of the present invention to provide a flexing hair clipper with a flexing cutting blade adjuster which adjusts automatically to the contours of a client's head to prevent the blade set getting stuck and causing cuts and irritation to the scalp.

(4) It is also an object of the present invention to provide a hair clippers device with infinitely variable blade distances from the scalp of the patron.

(5) Other objects which become apparent from the following description of the present invention.

SUMMARY OF THE INVENTION

(6) The hair clippers of this invention use a self-contained motor-driven adjustment mechanism to adjust the relative position of the stationary and reciprocating blades of a common type of blade set, preferably with a flexing blade set to adjust to the contours of the scalp of the customer having his or her hair being cut and trimmed.

(7) While other on/off switches can be used, preferably two momentary switches operable by the thumb of the hand holding the clipper afford a barber total automatic adjustment with the clipper itself in an on or off condition. There is no need for two-handed fidgeting or selection of only a few discrete increments of length adjustment as with the commonly available models. Since the small gear motors used for the adjustment are brush type or brushless permanent magnet motors which are operated by direct current, the adjustment feature is most compatible with cordless clippers already using an on-board DC source in the form of a re-chargeable battery to drive the reciprocating blade. The invention will be described as a modification of a cordless clipper, although AC driven corded type clippers can also be modified with this feature by the addition of an on-board AC to DC power supply for the adjustment motor.

(8) In the first embodiment, a modified blade set is used such that a gear rack is attached to the stationary blade. It is engaged with a worm gear pinion driven by a low-speed gear motor through a reversible drive circuit. Either limit switches, limit sensors, or over-current sensors are used to disable the adjustment motor at either the long or short hair end limits. The motor then can only be driven in the opposite direction.

(9) In the second embodiment, a conventional blade set is used. The modification is such that a

motor-driven final gear replaces the manual handle thereby retaining the original mechanism (of any type) that is used to move the stationary blade relative to the reciprocating blade in the conventional blade set. A timing belt bushing couples a rear mounted adjustment motor to a front side-mounted gear train coupled to the shaft of the blade shifting mechanism. Attached to the timing belt bushing for linear back and forth excursions is a magnet with a pointer. The magnet is used to operate two normally closed magnetic reed switches placed at the opposite distal ends of the permissible excursion thereby serving the limit switch function. The pointer moves over a tri-colored linear scale viewable by the barber from the top of the hair clipper; this quickly indicates the hair length setting. A plastic housing cover over the adjustment motor at the back and over the timing belt bushing and gear train at the side encloses the entire compact mechanism.

(10) In a third embodiment, a flex clipper is described which, in addition to the aforementioned powered hair cutting length adjustment feature, provides an additional feature to help the cutting blade set float more effortlessly, by adjusting automatically to the contours of a client's head to prevent getting stuck and causing cuts and irritation to the scalp.

(11) To achieve this automatic adjustment, the blade set with motor driven length adjuster in now housed in a separate module. Compliance is introduced between this module and the main housing of the flex clipper. The blade set can now tilt a small amount in any direction to automatically adjust to the local scalp contours while the cutting process is controlled as usual by grasping the main housing. The rigid attachment of the blade set to the housing is replaced by a flexing compliant attachment. Two methods are described, one is by using a large diameter short bellows while the other method uses a short length (a ring) of thick-walled elastomeric foam tubing which provides a similar function.

(12) Both flexing compliant attachments permit tilting and a small amount of linear axial movement between blade set and main housing, but both resist any relative rotational movement between blade set and main housing. This rotational resistance ensures good control of the blade set by keeping the cutting edge always aligned with the top surface of the housing (as in a normal rigid attachment) except for any minor local tilting. This rotational stiffness must also resist the driving torque of the motor driving the reciprocating cutter blade.

(13) Since the drive motor for the reciprocating cutter blade is in the main housing and the crank mechanism and blade set are in a separate module, a flexing compliant motor coupling that can follow any blade movements relative to the main housing is required. A metal bellows coupling of a diameter which fits inside the hollow interior of coupling bellows or foam ring is used. To keep the mass and size of the forward blade set module low, a modified cutting length adjuster mechanism is used; for example, in one embodiment, it uses a miniature stepper motor with a lead screw. The powering and control cable from the stepper motor driver in the main housing is also guided through the hollow interior of the coupling member.

(14) The flexing compliance (i.e., spring characteristics) of the coupling member as well as the damping characteristics can be determined by the geometric design and material selected. The proper “feel” can be achieved through simulation and actual prototype testing known to those skilled in the art of hair clippers technology. While the damping characteristics are not as important as the compliance, they determine the smoothness and sound deadening performance. For the bellows, a wide variety of thermoplastic elastomers (TPE's) or rubbers can be used. By using thin material crossection, even normally rigid plastics such as nylons or polypropylene can be used. Geometric design of the bellows includes overall length and diameter as well as number and shape of convolutions. By using filled TPE's or alloys of rubber/TPE a wide variety of damping characteristics can be designed in. Foamed rubbers or TPE's can be used for a foam ring coupling; other parameters that can be selected include type of cell (open or closed) and size of the cells. Material selection must also pay attention to longevity and compatibility with lubricants and hair conditioners.

(15) In a further alternate embodiment, as shown in FIGS. **14-23**D, the blade set is also capable of

flexing around the contours of the customer's hair, scalp and skull. Power is applied to the blade set by a conventional motor within the handset housing of the clippers. The motor may be activated by conventional tap switches, rotating wheel switches, or other manually activated switches. Instead of a flexible cylindrical neck, as in the aforementioned flexing embodiment, in this embodiment, the blade set is pivotable upward from a first position to a second position, whereby the blade set is controlled by a semi-rigid flexible belt bushing piece, i.e. known as a “flexor” which is positioned on the bottom of the clippers housing and which includes a semi-rigid flexible curved distal end which biases against a portion of the blade set to urge the blade set to move around the contours of the customer's hair, scalp and skull during the process of a hair cutting. The flexor counteracts the propensity of the upwardly pivoted blade set to pivot outwardly and holds the blade set in a mid-point position so that the blade set can push in or push out while moving over the three-dimensional curvature of the hair, scalp and skull of the customer. The flexor gently pushes the pivoted blade set to a flexing motion or a relaxed motion against the hair, scalp and skull of the customer. The moving blade of the blade set moves horizontally against the stationery blade of the blade set during the cutting of the hair. A spring is provided to hold the movable blade and the stationery blade closely adjacent and parallel to each other. The spring is located under a driver which has a driver bracket attached to the blade set. The driver moves the movable blade against the stationery blade to facilitate cutting of the hair on the scalp and skull of the customer. The movable blade is moved closely adjacent and parallel to the stationery blade, by an eccentric rotating cam which is powered by the motor (such as for example the motor M in the other embodiment shown in drawing FIG. **8**) inside the housing of the hair clippers. The eccentric cam causes the driver to move the movable blade of the blade set adjacent to the stationery blade in subsequent left and right sets of multiple parallel movements, during cutting of the hair of the scalp and skull of the customer. The rod of the rotating cam is positioned between an open U-shaped driver and causes the movement of the movable blade against the adjacent surface of the stationery blade. The U-shaped driver is mounted to a driver bracket which is attached to the blade set, and urges the blade set forward or back in an infinitely variable range of motion, limited by the pushing or release of the flexor against the pivoted blade set. A pivot mount is attached to a pivot plate which pivots the blade set and pivots about a pin, which connects the movable pivot plate and blade set to the stationery pivot mount, which is mounted to a base. The base is preferably connected to a three-sided shroud which covers the two sides and front of the pivoting mechanisms in front of the pivotable blade set.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present invention can best be understood in connection with the accompanying drawings. It is noted that the invention is not limited to the precise embodiments shown in drawings, in which:
(2) FIG. **1** is a perspective view of a typical prior art hair clipper with manual adjustment lever at the side.
(3) FIG. **2** is a side elevation of the prior art hair clippers of FIG. **1**.
(4) FIG. **3** is a side elevation of a motor-driven mechanism for adjusting the stationary blade of a clipper blade set showing a rack and worm gear pinion of the first embodiment.
(5) FIG. **4** is a perspective view of the hair clipper of this invention incorporating the mechanism of FIG. **3**.
(6) FIG. **4**A is a side view in crossection of the hair clipper of this invention, showing the primary motor therein.
(7) FIG. **5** is a wiring diagram of the adjustment motor using an “H-bridge” type of reversible driver.

(8) FIG. **6** is a top view of a second embodiment hair clipper with motor-driven adjustment of this invention.
(9) FIG. **7** is a side elevation of the second embodiment clipper with the housing cover removed to reveal the timing belt bushing and gear train mechanism.
(10) FIG. **8** is a side exploded elevation of the flex clipper embodiment of this invention.
(11) FIG. **9** is an assembled perspective view of the flex clipper of FIG. **8**.
(12) FIG. **10** is a side elevation of a compliant coupling between blade set module and main housing based on the use of an elastomeric foam ring.
(13) FIG. **11** is a perspective view of an elastomeric foam ring.
(14) FIG. **12** is a perspective view of the cutting length adjuster mechanism as attached to the adjustable comb plate.
(15) FIG. **13** is a high-level schematic diagram of the electrical elements of the flex clipper of this invention.
(16) FIG. **14** is a perspective view of an alternate embodiment for a hair clipper with a flexing pivoting blade set.
(17) FIG. **15** is a side view thereof.
(18) FIG. **16** is a front view thereof.
(19) FIG. **17** is an exploded view of the motor, eccentric cam, driver, pivot set, and blade set thereof.
(20) FIG. **18** is a close-up perspective view of the flexing bushing belt flexor associated with the pivoting blade set.
(21) FIGS. **19**A, **19**B, **19**C, **19**D are top, perspective, side and end views of the driver bracket.
(22) FIGS. **20**A, **20**B, **20**C, **20**D are top, perspective, side and end views of the pivot plate.
(23) FIGS. **21**A, **21**B, **21**C, **21**D are top, perspective, side and end views of the pivot mount.
(24) FIGS. **22**A, **22**B, **22**C, **22**D are top, perspective, side and end views of the driver.
(25) FIGS. **23**A, **23**B, **23**C, **23**D are top, perspective, side and end views of the inner cover/shroud.

DETAILED DESCRIPTION OF THE INVENTION

(26) FIGS. **1** and **2** show two views of a conventional cordless electric hair clipper **1** with on/off switch **3**, conventional blade set **2**, and side manual incremental adjusting handle **4**. The detents **5** engage handle **4** to set the minimum hair cutting length at one of the selections.
(27) FIG. **3** shows the mechanism which uses gear motor **10** driving worm gear pinion **11** to perform an adjustment of stationary blade **14** relative to reciprocating blade **13** in blade set **12**. A gear rack **15** subassembly is attached to blade **14** and engages pinion **11**. Also shown in this view are limit switches **16** and **17** at the longest and shortest settings respectively. FIGS. **4** and **4**A show clipper housing **20** with the adjustment feature. Conventional on/off switch **25** connected to clipper motor **24** (shown schematically as an encircled “M”) is at one side while momentary (or “tap”) switches **21** and **22** on the top surface are used to energize gearmotor **10** in a direction toward longer settings or shorter settings respectively. Gearmotor **10** is enclosed in descending housing **26**, which descends below clipper housing **20**. While FIGS. **3**, **4** and **4**A show a worm gear, it is anticipated that other gears may be used, such as rack and pinion gears or other gears known to those skilled in the art.
(28) FIG. **5** is a wiring diagram for the first embodiment of FIGS. **3** and **4** wherein gearmotor **10** is a simple brush type permanent magnet type driven by a common “H-bridge” drive module **35**. Battery **30** is used primarily to power clipper motor **24** through on/off switch **25**. It is also used as the power source for the adjustment feature. Drive module **35** has two direction inputs for clockwise and counterclockwise operation, an “ON” input, and power input and motor output connections as shown. In operation, if normally open switch **22** is pushed, a signal will flow through normally closed limit switch **17** energizing the ON input through isolation diode **36**; motor **10** will be driven clockwise until either switch **22** is released or limit switch **17** is opened at the end of the excursion. Similarly, if switch **21** is pushed, counterclockwise operation is achieved through

limit switch **16** and isolation diode **37**. Once a limit switch is opened, motor **10** can only be driven in the opposite direction until the open limit switch is again closed.

(29) FIGS. **6** and **7** show top and side views of the second embodiment of motor-driven minimum hair length adjustable hair clippers. The same circuit shown in FIG. **5** is completely applicable to this embodiment as well. The same momentary ("tap") switches **21** and **22** are used to control motor **10** which is now placed at the back end of hair clipper **40**. Except for the addition of switches **21** and **22**, the housing **41** and internal mechanism is identical to that of the prior art cordless clipper shown in FIGS. **1** and **2**. In this embodiment, a conventional blade set **12** and internal blade adjusting mechanism is used. The feature of this embodiment couples through the shaft formerly engaged with a manual handle **4**. This is shown at the center of output gear **51**. In the top view of FIG. **6**, housing cover **42** is a plastic shell used to enclose the feature mechanism. In FIG. **7**, this cover **42** is removed to reveal the mechanism; the position is shown in dashed lines. On the top edge of cover **42** is a tri-colored strip **43** with green region **45** denoting the long settings, yellow region **46** denoting medium length settings, and red region **47** denoting short settings. This scale is meant to be read relative to the position of pointer assembly **44** which is attached to timing belt bushing **55** transmitting power and torque from pulley **57** mounted on motor **10** to pulley **56** attached to the input gear of gear train **50**.

(30) Gear train **50** is used to adjust the torque at output gear **51** and to match the speed and torque of gear motor **10** and the desired indicating excursion of belt bushing **55** so as to form an ergonomic range. Besides the pointer on top, pointer assembly **44** also carries a small powerful magnet to operate limit switches **16** and **17** which are now implemented as normally closed magnetic reed switches. On/off switch **25** fits between timing belt bushing **55** and pokes through a side switch hole in housing cover **42**. While FIGS. **6** and **7** show a particular embodiment for an exterior mounted embodiment, it is anticipated that other exterior mounted embodiments may be used, such as those known to those skilled in the art.

(31) While this third embodiment will be described as for a flex hair clipper with both powered hair cutting length adjustment as well as flexing compliance introduced between the main housing and blade set module, it should be noted that the flexing compliance feature to permit the blade set to automatically adjust to scalp contours and irregularities can be afforded to hair clippers without the powered hair cutting length adjustment. If the latter feature is not implemented, the blade set module will just contain the blade set and crank mechanism with coupling to the drive motor in the main housing which operates the reciprocating cutting blade; there would not be a cutting length adjustor motor, adjuster mechanism attached to the comb plate, nor a housing for the adjuster motor.

(32) FIG. **8** shows an exploded view of the major components of this embodiment. Flex clipper **100** has main housing **110** which contains drive motor **116** with shaft **112** which drives the reciprocating cutter blade **113**, rechargeable battery **135** (unless it is an AC driven corded model), and an electronic driver module **160** for the hair cutting length adjuster motor **145** located in blade set module **140** at the left of the FIG. **8**. Rigid coupling ring **118** is attached at the coupling end of housing **110**. Blade set module **140** carries adjustable comb plate **114**, reciprocating cutter blade **113**, internal crank mechanism **143** for reciprocating cutter blade **113**, drive shaft **142** for crank mechanism **143**, housing **144** for internal hair length adjustment motor **145**, internal hair length adjustment direct comb plate mechanism **114** (shown in FIG. **12**), and a rigid coupling ring **146**.

(33) Also shown in FIG. **8** is molded compliant bellows **120** with integral mounting rings **126** and **128** is shown between blade module **140** and main housing **110**, which it couples together. Metal bellows **130** couples drive motor **116** in main housing **110** and crank drive shaft **142** in blade module **140**. Cable **148** powers and controls motor **145** for hair cutting length adjustment from electronic step driver module **160** contained in housing **144**; it is passed through the hollow interior of bellows **120**.

(34) FIG. **9** shows an assembled flex clipper **100** showing tap switches **21** and **22** for adjusting

cutting length and clipper operating switch **25**. A thumb operable reverse direction wheel **23** can also be optionally used. Bellows **120** is shown coupling blade module **140** to housing **110** in a flexing compliant fashion. The length of bellows **120** as shown in FIGS. **8** and **9** may be shorter than shown based on the design and materials of the bellows. Bellows integral collars **126** and **120** fit over fixed collars **146** and **118** on blade module **140** and housing **110** respectively. Fasteners, such as self tapping screws, are used to secure the bellows collars to collars **146** and **118** which preferably have transverse holes in registration.

(35) FIG. **10** shows an alternate embodiment of an assembly **150** of resilient foam ring **152** with attached metal collars **154**, which are adhesively attached or vulcanized as appropriate to the collar material. The assembly **150** of FIG. **10** can be used in lieu of custom molded bellows **120**. Depending on many variables known to those skilled in the hair clippers technology, such as desirable product life, product price point, manufacturing cost, performance, volume, and materials used, either the bellows or the foam ring assembly **150** may be the better choice.

(36) FIG. **11** shows a perspective view of the foam ring prior to attachment of coupling rings **154**.

(37) Although other types of flexing compliant motor couplings can be used, such as a variety of spring type couplings, the preferred coupling between shaft **112** and shaft **142** for reciprocating blade drive is a metal bellows coupling **130** such as those supplied by Servometer of Cedar Grove, NJ. This type of coupling easily fits inside the hollow bellows **120** or foam ring **152** central hole while not interfering with the degrees of freedom of the bellows or foam ring.

(38) FIG. **12** shows the simple direct comb plate **114** adjustment mechanism which includes preferably stepper motor **145**, and a fastening mechanism, such as, for example, threaded bracket **149** and fine lead screw **147**. Although other methods can be incorporated, a stepper motor **145** is preferred to a DC gearmotor due to size and complexity. At about 6 mm diameter and 9.5 mm long, a FDM0620 stepper motor from Micromo of Clearwater, FL is very compact and is driven with 20 steps per revolution to drive lead screw **147**.

(39) FIG. **13** shows a schematic diagram for the flex clipper. It is noted that no limit switches are required because step motors can just “lose steps” with no damage when a hard stop is encountered. Tap switches **22** and **24** determine the direction of rotation of stepper motor **145** by supplying the proper sequence of steps from step driver module **160** over cable **148**. Reciprocating blade motor **116** for reciprocating cutter blade **113** is directly powered through switch **25**. Battery **135** (or equivalent DC power supply for corded versions) supplies power to both reciprocating blade motor **116**, and to stepper motor **145**, through step driver module **160**.

(40) FIGS. **14-23**D show a further flexing hair clipper **200** with an external handle body housing **200***a* having an interior handset housing **208** therein. In this further alternate embodiment, the blade set **206**, **207** is also capable of flexing around the contours of the customer's hair, scalp and skull. As shown in FIG. **17**, power is applied to the blade set **206**, **207** by a conventional motor M (as in FIG. **17**) within the interior handset housing **208** of the hair clippers **200**. The motor M may be activated by conventional tap switches, rotating wheel switches, or other manually activated switches S. Instead of a flexible cylindrical neck, as in the aforementioned flexing embodiment of FIGS. **8-13**, in this embodiment, in FIGS. **14-23**D, the blade set **206**, **207** is pivotable upward, as shown in FIG. **18**, from a first lower position FPos, through a mid-point MPt, to a second raised position SPos, whereby, as also shown in FIG. **18**, the blade set **206**, **207** is optionally controlled by a semi-rigid flexible belt bushing piece **215**, i.e. known as a “flexor” **215** which is positioned on the bottom surface of the external handle body housing **200***a* of clippers **200**. The flexor **215** includes from top to bottom a series of joined, parallel rippled wave-like rounded peaks **215***c* and valleys **215***d*. The flexor **215** includes a semi-rigid flexible curved distal end **215***a*, attached to a linear proximal body portion **215***b*. The semi-rigid flexible curved distal end **215***a* biases against a portion of the blade set **206**, **207** to urge the blade set **206**, **207** to move around the contours of the customer's hair, scalp and skull during the process of a hair cutting. The flexor **215** counteracts the propensity of the upwardly pivoted blade set **206**, **207** to pivot outwardly and holds the blade set

206, **207** in a mid-point position so that the blade set **206**, **207** can push in or push out while moving over the three-dimensional curvature of the hair, scalp and skull of the customer. The semi-rigid flexible curved distal end **215***a* of the flexor **215** gently pushes the pivoted blade set **206**, **207** to a flexing motion or a relaxed motion against the hair, scalp and skull of the customer. The linear extending proximal body portion of the flexor **215** stabilizes the flexor **215** against the body of external handle body **200***a* of the hair clipper **200**.

(41) Although dimensions of the component parts of the hair clipper **200** may vary, FIGS. **19**A, **19**B, **19**C, **19**D, **20**A, **20**B, **20**C, **20**D, **21**A, **21**B, **21**C, **21**D, **22**A, **22**B, **22**C, **22**D, **23**A, **23**B, **23**C and **23**D each show preferred dimensions expressed in inches, or portions of inches, where the dimension areas indicate the decimal portions of an inch, including radii indicated with a capital "R" preceding each of the radii.

(42) The moving blade **206** of the blade set **206**, **207** moves horizontally against the stationery blade **207** of the blade set **206**, **207** during the cutting of the hair. As shown in FIG. **15**, a spring SP is provided to hold the movable blade **206** and the stationery blade **207** closely adjacent and parallel to each other. The spring SP is located under a driver **204** which has a driver bracket **201** attached to the blade set **206**, **207**. The driver **204** is connected to the driver bracket **201**, which pivots about driver pin **210** by the force of driver **204**. The driver **204** moves the movable blade **206** against the stationery blade **207** to facilitate cutting of the hair on the scalp and skull of the customer. The movable blade **206** is moved closely adjacent and parallel to the stationery blade **207**, by an eccentric rotating cam **211** which is powered by the motor M (such as for example the motor M in the other embodiment shown in drawing FIG. **8**) inside the interior handset housing **208** of the hair clippers **200**. Motor "M" is powered by a power source being one of a battery B within the interior handset housing **208** or an AC driven corded power source AC, (or equivalent DC power supply for corded versions) which supplies power to both reciprocating blade motor "M", or, as also shown in FIG. **8**, to a stepper motor **145**, through a step driver module **160**.

(43) The eccentric cam **211** causes the driver **204** to move the movable blade **206** of the blade set **206**, **207** adjacent to the stationery blade **207** in subsequent left and right sets of multiple parallel movements, during cutting of the hair of the scalp and skull of the customer. The rod **211***a* of the rotating eccentric cam **211** is positioned between the open U-shaped driver **204** and causes the movement of the movable blade **206** against the adjacent surface of the stationery blade **207**.

(44) The U-shaped driver **204** is mounted to a driver bracket **201** which is attached to the blade set **206**, **207**, and urges the blade set **206**, **207** forward or back in an infinitely variable range of motion, limited by the pushing or release of the bushing belt flexor **215** against the pivoted blade set **206**, **207**. A pivot mount **202** is attached to a pivot plate **205** which pivots the blade set **206**, **207** and pivots about a pin **209**, which connects the movable pivot plate **205** and blade set **206**, **207** to the stationery pivot mount **202**, which is mounted to a base. The interior handset housing **208**, supports motor "M" within, and the interior handset housing **208**, is preferably connected to external handle body housing **200***a*, as shown in FIG. **18**, which covers the two sides and front of the pivoting mechanisms of the hair clipper **200** in front of the pivotable blade set **206**, **207**, including U-shaped driver **204** communicating with rod **211***a* of the rotating eccentric cam **211**. Hair clipper **200** also optionally includes flexing bushing belt flexor **215** on the bottom fourth side of the external handle body housing **200***a* of the hair clipper **200**.

(45) In the foregoing description, certain terms and visual depictions are used to illustrate the preferred embodiment. However, no unnecessary limitations are to be construed by the terms used or illustrations depicted, beyond what is shown in the prior art, since the terms and illustrations are exemplary only, and are not meant to limit the scope of the present invention.

(46) It is further known that other modifications may be made to the present invention, without departing the scope of the invention, as noted in the appended Claims.

Claims

1. A flex hair clipper comprising: a manually graspable external handle body housing an internal housing having a motor with an eccentric cam; a driver in operative engagement with the eccentric cam; a blade set including a movable blade and a stationary blade, the movable blade being in operative engagement with the driver so that the movable blade moves back and forth against the stationary blade via operation of the eccentric cam of the motor; a pivot plate pivotally connecting the blade set to the handle body so that the blade set pivots along an arcuate path between an inward first position to an outward second position; and a semi-rigid flexor having a semi-rigid flexible curved distal end and a linear body portion defining a proximal end of the flexor, the semi-rigid flexible curved distal end is attached to a back side of the blade set, the linear proximal body portion is attached to a backside of the handle body, the flexor counteracts a propensity of the blade set to move to the outward second position so that the blade set is held in a middle position between the first position and the second position so that the blade set can pivot inward toward the first position or outward toward the second position in response to engaging contours of a user during use.
2. The flex hair clipper as in claim 1, the flexor includes a series of joined, parallel rippled peaks and valleys extending from the distal end of flexor to the proximal end of the flexor.
3. The flex hair clipper as in claim 1, further comprising the motor being powered by a power source being selected from a group consisting of a battery or an AC driven corded power source.
4. The flex hair clipper as in claim 1, further comprising a spring being provided to hold the movable blade and the stationary blade adjacent and parallel to each other, the spring being located adjacent to the driver, a driver bracket attached to the movable blade, and the driver is pivotally connected to the driver bracket via a driver pin to define the operative engagement between the movable blade and the driver.
5. The flex hair clipper as in claim 1, the handle body includes a stationary pivot mount, the pivot plate is pivotally connected to the stationary pivot mount via a pin to define the pivotal connection between the blade set and the handle body.